

Hybrid order fulfillment optimization

Fast feasible allocation with exact local improvement
and honest quantum evidence.

HiGHS baseline • MILP polish • constraint-aware QAOA neighborhoods

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Submission evidence

247 / 247

recorded result rows feasible

+197.2

objective gain on the coordinated-move trap

156 qubits

IBM hardware study; 18 QPU jobs

16 canonical manifests verified

The decision is a constrained allocation—not a single score

INPUT

Orders, products, plants

- demand and promised dates
- available-to-promise (ATP) inventory
- eligible fulfillment options
- transport, transfer, and lateness costs

DECISION

Choose at most one option per group

$$x_{ij} \in \{0,1\}$$

1 when option j is assigned to order group i ; 0 otherwise.

Unfilled demand is allowed, but penalized.

OBJECTIVE

Maximize service value

value – shipping – transfer
– lateness – unfilled penalty

Subject to

ATP limits • plant capacity • option eligibility •
one option per group

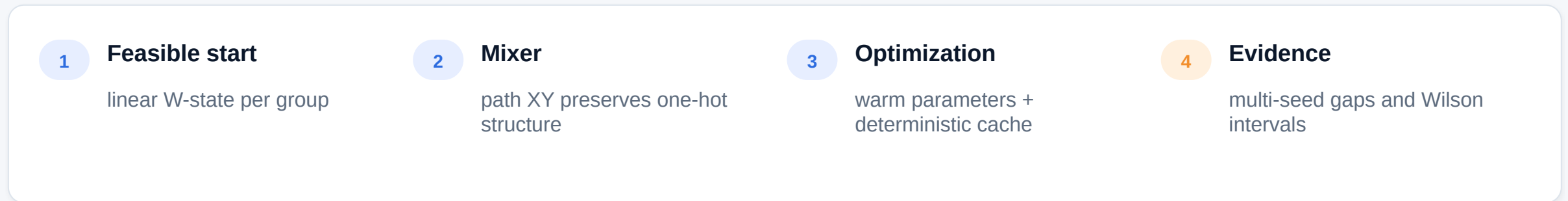
Why it is hard

Inventory couples many groups. Improving one allocation can require coordinated changes elsewhere—exactly the failure mode that local search and quantum neighborhoods target.

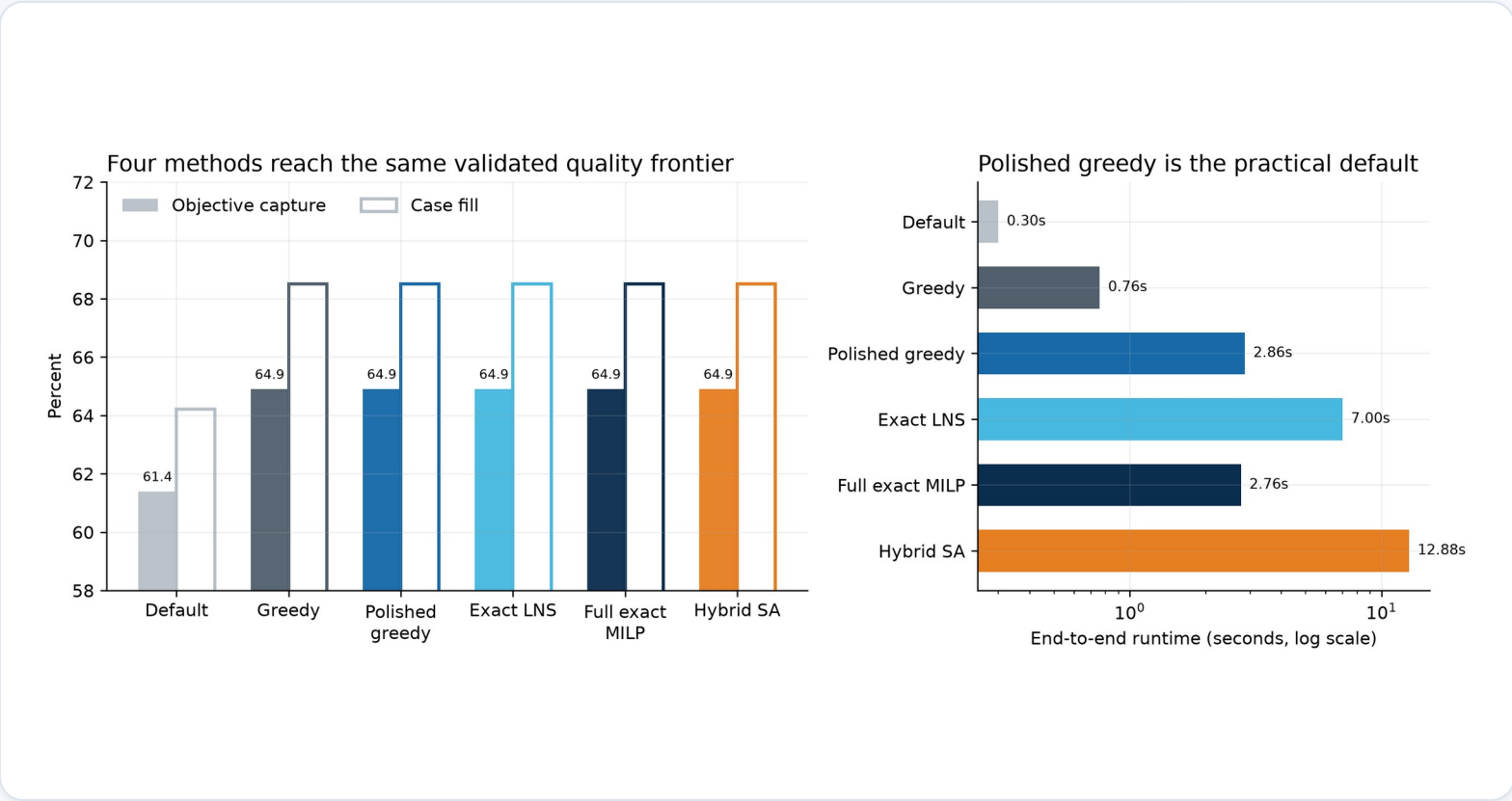
The hybrid solver keeps feasibility deterministic



Quantum neighborhood (optional)



On the real 20-group subset, polish reaches the exact objective



64.90%

objective capture after polishing

68.50%

demand fill rate

2.86 s

polished median runtime

0

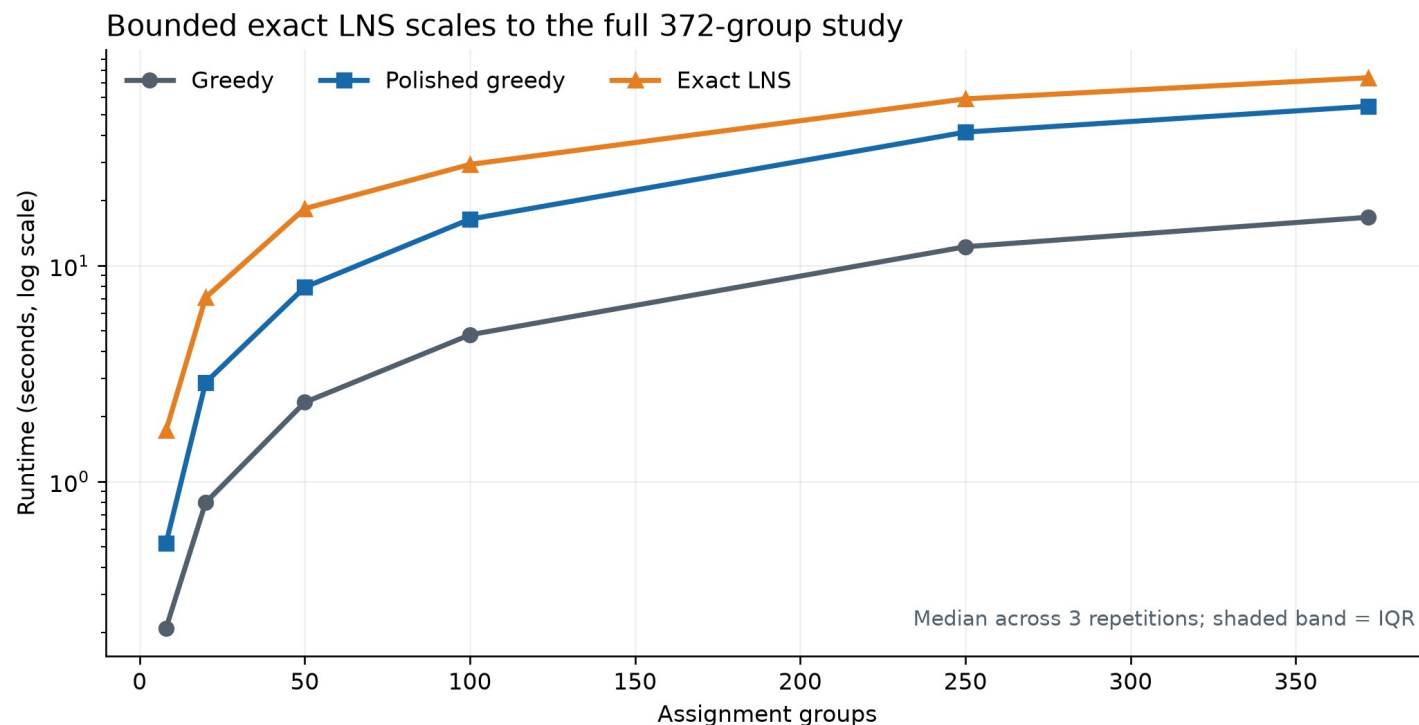
numeric residual violations

+197.2 on the coordinated trap

Exact LNS and the hybrid escape the greedy basin; this is the strongest evidence for coordinated neighborhoods.

Objective capture = achieved objective ÷ total requested merchandise value; it is not an optimality gap.

HiGHS stays portable; exact methods are escalation tools

**DEFAULT**

SciPy + HiGHS

Open-source, cross-platform, and always used unless another backend is explicitly requested.

OPTIONAL

Gurobi parity

A licensed benchmark path; never required and never selected silently.

ESCALATE

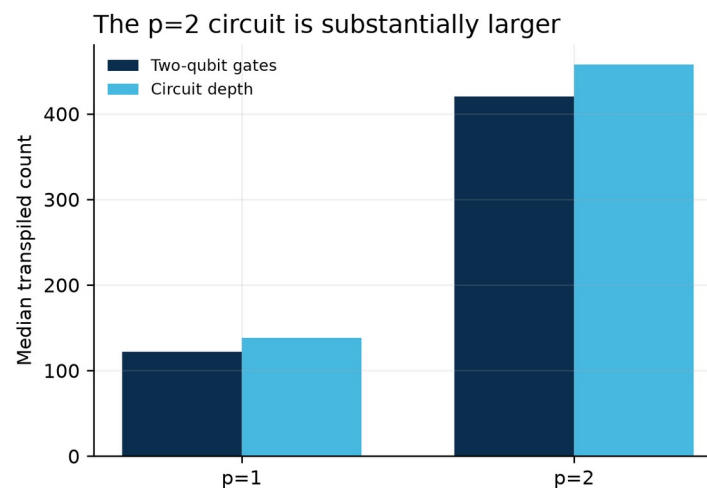
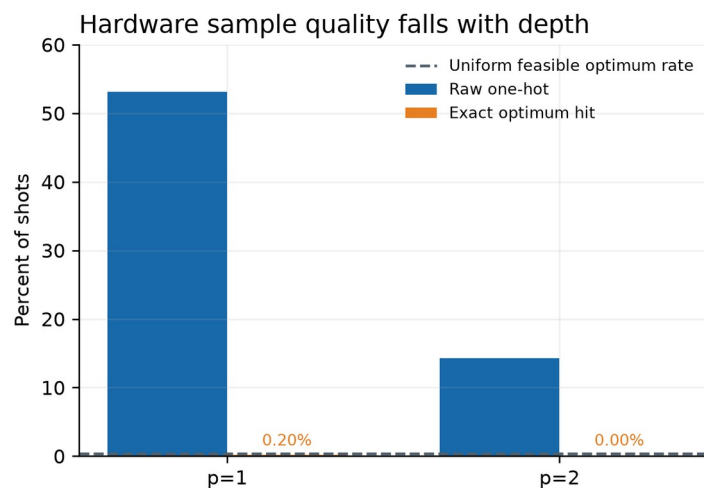
Exact LNS / quantum

Spend extra compute only on hard, coupled neighborhoods.

At 372 groups: greedy ≈ 16.7 s • polished ≈ 54.6 s • exact LNS ≈ 74.0 s (medians).

Hardware evidence favors shallow, constraint-aware circuits

IBM Marrakesh: 18 matched QPU jobs, 512 shots each



p = 1

122 two-qubit gates •
depth 138

50.8–59.0%

median raw one-hot
validity

p = 2

420 two-qubit gates •
depth 458

12.1–15.2%

median raw one-hot
validity

Interpretation

Deeper ideal QAOA improves, but hardware noise overwhelms the benefit. Recourse restores final feasibility.

No quantum-advantage claim: 18 QPU jobs are evidence of execution and failure modes, not superiority. Archived runs predate the strengthened W-state/path-mixer implementation.

Deploy the hybrid now; treat quantum as a measured R&D lane

RECOMMENDATION

Production path

- Use polished HiGHS as the portable default.
- Trigger exact LNS only when coupling or stagnation is detected.
- Benchmark Gurobi only when licensed; require objective parity.
- Log provenance, feasibility, gaps, and sampler quality on every run.

Improvement roadmap

NOW

Operational tuning

Scenario stress tests • calibrate penalties • automated hard-neighborhood detection

NEXT

Search efficiency

Parallel LNS • adaptive neighborhood size • incumbent diversity • warm starts

R&D

Quantum evidence

Hardware-aware topology • CVaR objectives • angle transfer • stronger mitigation

FEASIBLE FIRST

REPRODUCIBLE

QUANTUM CLAIMS: EVIDENCE ONLY

The strongest solver is the one that improves decisions without weakening trust.